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ORIGINAL RESEARCH

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The impact of nonoperative hip and core injuries on National Football League athlete performance

Joseph S. Tramer[a], Toufic R Jildeh[a], Joshua P Castle[a], Patrick Buckley*[a], Caden Nowak[b] and Kelechi R. Okoroha[c]

[a]Department of Orthopaedic Surgery, Henry Ford Hospital, Detroit, United States; [b]Wayne State University School of Medicine, Detroit, Michigan, United States; [c]Department of Orthopedic Surgery, Division of Sports Medicine, Mayo Clinic, Minneapolis, Minnesota, United States

ABSTRACT

Objectives: Hip and core injuries are common in National Football League (NFL) athletes; however, the impact following injury remains unclear. The goal of this manuscript was to determine the impact of nonoperative hip and core injuries on return to play and performance.

Methods: NFL athletes who sustained a hip or core injury treated nonoperatively between 2010 and 2016 were identified. Offensive and defensive power ratings were calculated for each player's injury season and two seasons before and after to assess longitudinal impact. A matched control group without an identified hip and/or core injury was assembled for comparison.

Results: A total of 41 offensive and 71 defensive players with nonoperative hip or core injury were analyzed. All athletes returned to play; offensive and defensive players missed 4.0 ± 5.2 and 3.1 ± 2.6 games after injury, respectively. Offensive players played fewer cumulative career games returning from core injury versus hip (23.5 ± 20.6 vs 41.0 ± 26.4). Defensive players played fewer games (58.1 ± 41.1 versus 37.4 ± 27.1, $p < 0.05$) with lower defensive power rating (133.9 ± 128.5 versus 219.8 ± 212.2, $p < 0.05$) cumulatively after hip or core injury. Additionally, 2 years following injury, defensive players played fewer games compared to controls (9.5 ± 7.0 versus 10.9 ± 6.8, $p < 0.05$). Following hip injury specifically, NFL defenders played fewer games (39.8 ± 27.9 vs 61.9 ± 38.8; $p < 0.05$) and had a lower defensive power rating (145.9 ± 131.7 vs 239.0 ± 205.9; $p < 0.05$) compared to before injury.

Conclusion: Overall, NFL players return to play following nonoperative hip and core injuries. Defensive players played in fewer games following hip or core injury compared to controls; offensive players were unaffected. Hip injuries have a greater impact on performance compared to core injuries in defensive athletes; offensive players played fewer games upon return from core injury.

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Muscle injuries; football; groin pain; hip/pelvis/thigh; core

Introduction

Injuries in the National Football League (NFL) have been shown to have a significant financial burden to both players and organizations [1,2]. A systematic review of athletes entering the NFL combine with a musculoskeletal injury that required orthopedic intervention are uniformly at risk for decreased performance and fewer games, the effects of which resonate throughout an NFL player's career [3,4]. Although knee and shoulder injury account for the highest incidence of time loss injuries among NFL players, they are at significant risk for injuries to other anatomic regions, particularly the hip and core [5–8].

Hip and core injuries in the NFL account for approximately 10% of NFL injuries; the majority are classified as muscle strains [9,10]. In fact, hip flexor strains occurred 268 times over a 10 year period in the NFL, and were the most common hip injury [10]. Football players are particularly vulnerable to core injuries such as rib fractures or contusions, back stiffness as well as abdominal/oblique strains, and athletic pubalgia due to the physical nature of the game [11,12]. Prior studies have examined the effect of surgical intervention on groin injuries, showing that operative intervention for athletic pubalgia led to significantly shorter career length and games per season compared to matched controls [13]; however, the impact of nonoperative hip and core injuries on NFL player performance has not been elucidated.

The purpose of our investigation was to determine the impact of nonoperative hip and core injuries on the performance in NFL athletes. We sought to determine if these injuries had a significant impact on return-to-play rates, as well as offensive and defensive player productivity. It was hypothesized that both offensive and defensive NFL players who sustained a nonoperative hip or core injury would return to play but play fewer games and experience worse performance following their return as compared to healthy controls matched for age, position, and NFL experience.

Methods

Data were collected retrospectively for this study; internal review board exemption was granted since all material is freely accessible. All NFL players who sustained a hip or core

CONTACT Joseph S. Tramer joe.tramer@gmail.com; Patrick Buckley pbucks89@gmail.com Department of Orthopaedic Surgery, Henry Ford Hospital, Detroit, Michigan,United States

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injury between 2010 and 2016 seasons were collected from publicly available sources such as injury reports, player profiles, team pages, and game summaries using methods previously utilized in other studies [13–16]. Indexed hip injuries included adductor strain, hip flexor strain, labral tears, hip inflammation, or impingement while core injuries included oblique, abdominal, or intercostal strains, as well as back spasms or stiffness. Injuries were cross-referenced with public websites including official team pages, official press releases, and personal player websites, using methodology that has been utilized by previous studies using publicly available databases [13–16]. The NFL mandates that teams quickly and accurately disseminate injury information to the league office, their opponents, local and national media, and the team's broadcast partners. Return-to-play (RTP) data was further confirmed by identifying gaps in reported statistics. Fractures, disc herniations, and any player who was reported to have undergone surgery for their injury were not included in the investigation.

Demographic data surrounding the injury were recorded such as BMI, age at injury, season in NFL that injury occurred, and position. The total number of games played before and after the reported injury date were also recorded. Performance data were collected for each player including total games played, total yards (passing, rushing, or receiving), touchdowns for offensive players; and total tackles, sacks and interceptions for defensive players. Offensive and defensive power ratings were also calculated from these variables, which have both been previously described and validated outcomes measures of a football player's overall contribution to the team's offense or defense [14,17,18]. They are calculated with the following formulas: Offensive power rating (OPR) = [(total yards/10) + (touchdowns/6)] Defensive power rating (DPR) = [total tackles + (total sacks × 2) + (total interceptions × 2)]

Overall career data as well as data in the 2 years prior to and following the date of injury were collected in order to better assess longitudinal changes in performance. Power ratings are intended to calculate productivity over a season, not individual games; therefore, the individual statistics were also evaluated on a per game basis to account for games missed. A control group was assembled by 2:1 matching injured athletes with healthy rostered NFL players who did not sustain a documented hip and/or groin injury by age, position, performance data, and NFL experience within three seasons in accordance with previously established methodology [13–15]. Control players were chosen based on an index date that was matched with the study player's date of injury. This control group was established to compare performance before and after hip/core injury. Additionally, if players did not participate in the season prior to the index injury season, they were excluded. Furthermore, athletes injured at the end of the season who were unable to return-to-play during the index season were excluded from the study. When calculating time missed from injury, bye weeks were not a factor, only the number of games missed. Power rating has been previously demonstrated in other investigations to be a valid and effective method for quantifying overall player performance [14]; however, since several studies demonstrated offensive power ratings are not an effective performance metric for offensive linemen (OL) or kickers (K), only offensive skill players (wide receivers [WR], running backs [RB], tight ends [TE], and quarterbacks [QB]) were included for offensive power ratings [17,18].

Statistical analysis

All continuous data are reported as mean ± standard deviation, while categorical data are reported as counts and column percentages. Comparisons between injured and control players are performed using independent 2-sample *t*-tests for continuous variables and using chi-square tests for categorical variables. Comparisons between pre- and post-data are performed using paired *t*-tests if the variable is normally distributed, and Wilcoxon signed rank tests if the variable is non-normally distributed. Statistical significance is set at $p < 0.05$. All analyses are performed using SAS 9.4 (SAS Institute Inc., Cary, NC, USA).

Results

A total of 112 (41 offensive and 71 defensive) NFL players were identified as having sustained a hip or core injury between 2010 and 2016 that also matched our inclusion criteria. The offensive group included 15 wide receivers, 10 offensive linemen, 6 tight ends, 5 running backs, 4 quarterbacks, and 1 kicker. Defensive players included 33 defensive backs, 19 linebackers, 10 defensive ends, and 9 defensive tackles. A 2:1 matched control group consisting of 224 players was compared to the study group and demonstrated no statistical differences in age, years of NFL experience, position, or BMI (Table 1). Pre-injury performance did not significantly differ between the injured group and healthy controls. No players sustained injuries requiring retirement.

All players sustaining a core or hip injury were able to return to play following their injuries. Eighty-three percent of offensive players and 86% of defenders were able to return to play in the same season as the injury, the remainder of the players were able to return in the following season. Injuries occurred mostly in wide receivers (37% of offensive injuries) and defensive backs (46% of defensive injuries) and there was a total of 81 hip injuries and 31 core injuries overall. Eleven (9.8%) of all injured players had documented re-injury during the study period. Offensive players missed 4.0 ± 5.2 games when suffering a hip or core injury; defensive players missed

Table 1. Offensive and defensive injury and control group characteristics.

	Offensive injury (*N* = 41)	Offensive control (*N* = 82)	*p*-Value	Defensive injury (*N* = 71)	Defensive control (*N* = 142)	*p*-Value
Mean age, y	25.7 ± 3.1	25.9 ± 2.9	0.71	26.8 ± 2.8	26.6 ± 2.8	0.65
BMI (kg/m^2)	30.6 ± 4.6	30.8 ± 4.7	0.79	30.0 ± 3.6	30.6 ± 4.4	0.33
Mean NFL experience (seasons)	6.9 ± 3.0	7.4 ± 3.2	0.41	7.3 ± 2.9	7.5 ± 2.3	0.65

Table 2. Cumulative career performance of offensive players before and after hip or core injury (N = 41).

	Before injury	After injury	Difference	p-Value
Games played	49.3 ± 40.1	35.0 ± 25.7	−14.3	0.06
Yards per game	37.6 ± 59.3	35.0 ± 62.2	−2.6	0.17
Touchdowns per game	0.27 ± 0.44	0.26 ± 0.49	−0.01	0.30
Power rating[a]	341.8 ± 550.6	205.7 ± 306.9	−136.2	0.03*

[a]Power Rating only includes offensive skill players (WR, RB, TE, QB; N = 30) [K and OL excluded].
*Denotes statistically significant difference (p <0 .05).

3.1 ± 2.6 games. Players with hip injuries missed of 2.9 ± 2.3 games; athletes with core injuries missed 4.8 ± 5.8 games.

Regarding performance, offensive skilled players' experienced a decrease in career power ratings following a hip or core injury compared to pre-injury (341.8 ± 550.6 vs 205.7 ± 306.9, p = .03) (Table 2) while control players demonstrated no difference (350 ± 708.9 vs 335 ± 680.5; p = 0.38). There were no significant differences in performance between the injured offensive skill group and healthy matched controls in the first 2 years after injury or cumulative post-injury performance (Table 3).

Defensive players experienced a significant decrease in games played (58.1 ± 41.1 vs 37.4 ± 27.1, p < 0.01) and power rating (219.8 ± 212.2 vs. 133.9 ± 128.8; p < 0.01) compared to preinjury (Table 4). When comparing to healthy controls, defensive players participated in significantly fewer games at 2 years post injury (9.5 ± 7.0 vs 10.9 ± 6.8, p < 0.05) while no significant difference was found for cumulative career games played (37.5 ± 26.9 vs. 41.4 ± 25.5, p = 0.16). Power ratings did not significantly differ between injured and control defensive players at 2 years or for cumulative performance following injury (Table 5).

Games played and performance analysis were further stratified by either hip or core injury. Following hip injury, defensive athletes played in fewer games (39.8 ± 27.9 vs 61.9 ± 38.8; p < 0.05) and had a lower defensive rating (145.9 ± 131.7 vs 239.0 ± 205.9; p < 0.05) compared to their career before injury. Conversely, offensive skilled players experienced a significant decrease in power rating compared to their career before injury (395.4 ± 567.4 vs 190.1 ± 327.6; p < 0.05). When compared to controls, both offensive skilled players and defensive players with either hip or core injuries did not experience a significant decrease in cumulative career power ratings following injury.

Discussion

The present investigation demonstrated a 100% return-to-play rate after hip or core injury for NFL players. Defensive players with either a hip or core injury demonstrated significantly fewer games played and both offensive and defensive players experienced decreased power ratings post-injury compared to before injury. Compared to healthy controls, defensive athletes played in significantly fewer games in the 2 years following injury but returned to near baseline at long-term career follow-up. Defensive players who sustained hip injuries alone had significantly decreased games played and power ratings in their post-injury careers compared to prior. Offensive players who sustained a core injury had significantly fewer

Table 3. Performance of offensive players who sustained a hip or core injury (N = 41) and healthy matched controls (N = 82) in both the first 2 years and cumulative career statistics following hip or core injury.

	Two years following injury			Cumulative post-injury performance		
	Injured	Controls	p-value	Injured	Controls	p-alue
Games played	17.8 ± 10.6	21.4 ± 12.2	0.11	35.0 ± 25.7	44.5 ± 30.3	0.09
Yards per game	41.8 ± 66.9	47.7 ± 78.3	0.36	35.0 ± 62.2	40.6 ± 70.8	0.33
Touchdowns per game	0.32 ± 0.54	0.35 ± 0.64	0.43	0.26 ± 0.49	0.29 ± 0.50	0.37
Power rating[a]	106.7 ± 178.5	143.5 ± 233.3	0.21	205.7 ± 306.9	335.8 ± 680.5	0.11

[a]Power Rating only includes offensive skill players (WR, RB, TE, QB; N = 30 injured; N = 60 controls) [K and OL excluded].

Table 4. Cumulative career performance of defensive players before and after hip or core injury (N = 71).

	Before injury	After injury	Difference	p-Value
Games played	58.1 ± 41.1	37.4 ± 27.1	−20.7	<0.01*
Tackles per game	3.1 ± 1.8	2.9 ± 1.6	−0.2	0.13
Sacks per game	0.10 ± 0.15	0.09 ± 0.18	−0.01	0.40
Interceptions per game	0.05 ± 0.07	0.05 ± 0.07	0.0	0.26
Power rating	219.8 ± 212.2	133.9 ± 128.5	−85.9	<0.01*

*Denotes statistically significant difference (p < 0.05).

Table 5. Performance of defensive players who sustained a hip or core injury (N = 71) and healthy matched controls (N = 142) in both the first 2 years and cumulative career statistics following hip or core injury.

	Two years following injury			Cumulative post-injury performance		
	Injured	Controls	p-Value	Injured	Controls	p-Value
Games played	9.5 ± 7.0	10.9 ± 6.8	0.04*	37.5 ± 26.9	41.4 ± 25.5	0.16
Tackles per game	3.4 ± 2.0	3.2 ± 2.1	0.29	2.9 ± 1.6	3.0 ± 1.7	0.29
Sacks per game	0.12 ± 0.24	0.12 ± 0.20	0.48	0.09 ± 0.18	0.10 ± 0.16	0.37
Interceptions per game	0.05 ± 0.09	0.04 ± 0.08	0.20	0.05 ± 0.07	0.05 ± 0.07	0.29
Power rating	68.3 ± 63.9	81.9 ± 62.5	0.07	141.1 ± 108.9	148.4 ± 130.5	0.22

*Denotes statistically significant difference (p < 0.05).

games played in their cumulative career following injury compared to healthy controls.

Hip and groin injuries occur frequently in football players and are a leading cause of injury and lost playing time during both practice and games [19]. In a descriptive epidemiologic study of 23,806 NFL injuries from 1997 to 2006, Feeley et al. found that injuries around the hip comprised 3.1% of total injuries, with muscle strains being the most common. Wide receivers (39.6% of offensive injuries) and defensive backs (47.1% of defensive injuries) were among the most affected by these injuries [10]. The present study found a nearly identical incidence of injury, with wide receivers accounting for 37% of offensive injuries and defensive backs 46% of defensive injuries, with all athletes able to RTP.

High rates of hip and groin injury have also been documented in National Collegiate Athletic Association (NCAA) football players. Kerbel et al. reviewed the incidence of hip and groin injuries across multiple sports and found that football players sustained 62.7 hip or groin injuries per 100,000 athlete exposures, with the majority of reported injuries to either the adductor or hip flexor musculature [20]. Specifically investigating muscular strains about the hip, Eckard et al. reviewed groin injuries in NCAA athletes and found that hip flexor and adductor muscle strains occurred at a rate of 1.93 and 1.43 per 10,000 athletic exposures in football players, respectively [21]. The present cohort similarly revealed groin or hip flexor strain/tear as among the most common injury type, comprising 56.1% of offensive and 67.6% of defensive injuries. Eckard et al. also found 7.4% and 6.9% recurrence rate of hip flexor and adductor strains during the study period, compared to 9.8% of all athletes in our cohort with a documented re-injury [21]. Adductor strains, specifically, have the potential to alter the mechanics of the anterior musculature about the hip, leading to increased risk of future rectus strains or athletic pubalgia [22]. Adductor injuries have been shown to cause significant increases in missed time from sport in other professional leagues. Long-term performance and career longevity, however, were not altered [23]. These findings suggest that although muscular strains may appear benign, athletes with these injuries are at risk for future strains and time away from sport. Considering groin injuries cause significant time missed in the NFL and other professional sports, prevention is worth investigating. A study by Tyler et al. revealed a dedicated physical therapy program targeted at adductor strengthening was beneficial in decreasing the incidence of groin strains in National Hockey League players; a similar training regimen may be beneficial in the NFL as well [24].

Little is known about the effect of nonoperative core injuries such as back spasm, core stiffness, and oblique strain on NFL players. A study utilizing a convenience sample of 267 NCAA football players with lumbar spine injuries found that 59.6% of athletes reporting muscular low back pain or contusion were able to return to sport in less than 24 h and 72.6% of players who did not return in <24 h returned within a week of injury [25]. Rebolledo et al. examined 214 NFL athletes at the combine and found that 50% had a history of abdominal core or lower extremity muscular injury, which was also associated with lower levels of vitamin D [9]. The effect of back and core injury has also been examined in other professional sports, as Camp et al. found that core and back injury accounted for 12% of injuries resulting in missed time in Major League Baseball (MLB) [26]. Likewise, Marshall et al. found that core injuries in MLB pitchers lead to 47.0 ± 5.6 days on the injured list with fewer strikeouts per nine innings in the year of injury [16]. In the our study, athletes sustaining a core injury missed an average of 4.8 ± 5.9 games following these injuries and all athletes were able to RTP without significant performance detriment. Additionally, offensive players who suffered a core injury played significantly fewer games throughout the remainder of their post-injury career compared to healthy controls, which could be partially attributed to similarities in athletic demand between a quarterback and MLB pitcher. While return-to-play rates are high and the current literature does not report significant changes in football performance, Domb et al. reviewed disability claims in retired NFL athletes and found that 63 of 65 former players reported symptomatic lumbar spine strain [27]. It is important to note that, while core, and more specifically lumbar spine strains may not be associated with any surgical pathology, the repetitive stress can have implications on the future health of the athlete.

Specifically evaluating hip injuries in the NFL, Feeley et al. noted that athletes lost an average of 12.3 days, with a median loss time of 4 days following a variety of hip injuries, such as muscle strains, contusions, and intra-articular pathology [10]. In the aforementioned study by Marshall et al., pitchers with hip injuries spent an average of 37.7 ± 8.1 days on the injured list and starting pitchers suffered from decreased fastball velocity during their initial season following injury [16]. Additionally, injuries around the hip joint have also been shown to lead to missed time in professional basketball, as Jackson et al. reviewed 24 years of data and found that hip injury accounted for 14.6% of all athletic injuries and lead to an average of 6.3 ± 10.2 days missed [28]. In our investigation, defensive players were more affected by hip injury than core injuries, with individual decreases in DPR following a hip injury but not a core injury, as well as fewer games played in the 2 years following injury compared to healthy controls.

Although both offensive skilled players and defensive players saw a decrease in power ratings compared to their own previous performance, neither group experienced a significant decline in performance compared to the control group following a hip or core injury. In a 2017 review of player injuries at the NFL combine, Beaulieu-Jones et al. demonstrated that among NFL athletes entering the draft, defensive players with an injury history played fewer games in their first 2 years in the league compared to healthy controls (22.2 ± 7.9 vs 24.2 ± 7.6 games; $p = 0.001$). Similarly, our study demonstrated that defensive players who suffered a hip or core injury played in fewer games the 2 years following compared to healthy defensive controls. Additionally, a systematic review by Wang et al. looked at 32 studies pertaining to injury surveillance and monitoring at the NFL combine and found that overall, defensive linemen and defensive backs appeared to have their game participation most affected by an injury history compared to offensive athletes [3]. This is likely due to the inherently violent nature of defensive play as compared to offensive play, which predisposes defensive

players to more energy transfer and higher impact, and thus higher injury rate. The defensive back position involves more efficient energy transfer in the kinetic chain during movements like backpedaling, quick reactionary changes in direction, and regular high-speed contact, which may help to explain the higher incidence of hip injuries among defensive players. However, it is important to note that in the present investigation performance decline following an injury was not significantly different for offensive skilled or defensive players when comparing to matched controls; thus, while athletes may expect some decrease in individual performance, it will not statistically exceed other NFL players of similar age and position.

Our study is not without limitations. Player injury reports were collected from publicly available internet sources, which may be subject to inaccuracies. However, the NFL mandates that teams provide accurate and prompt injury information to the league as well as the media. Similar to previous investigations, injury reports were confirmed with additional publicly available injury reports [13–15]. With pressure to stay in the game and continue playing, these injuries may also be unreported, as players might not disclose injuries to the team, medical staff, and media. Furthermore, while game statistics are an important indicator of player performance, there are many unmeasured contributions not captured; thus, true 'performance' of a player cannot be completely summarized by these statistics alone. Our use of player power rating attempts to mitigate this issue by generating a weighted value that emphasizes commonly agreed upon significant performance variables [17]. These power ratings, however, do not capture kicker or offensive linemen contributions; therefore, these players were excluded from power rating analysis. In addition, hip and core injuries include a broad spectrum of pathologies, and public injury reports often do not indicate the specific diagnosis. We are unable to access clinical exam findings, subjective reports from the athlete, imaging results, and attempted conservative treatment modalities. Future studies should aim to identify specific features that favor successful nonoperative management of common hip and core injuries and correlate with performance.

Conclusion

Overall, 100% of NFL players in our study returned to play following nonoperative hip and core injuries. Defensive players were more affected post-injury compared to offensive players. Hip injuries have a greater impact on performance compared to core injuries in defensive athletes; offensive athletes played fewer games upon return from core injury but were able to maintain their performance. At long-term follow-up, both offensive and defensive players were able to return to near baseline games played and power ratings.

ORCID

Patrick Buckley http://orcid.org/0000-0003-0650-929X

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